Food Delivery Analytics Project

Project Charter & Stakeholder Reference Document

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This document is the single reference for the project's business context: who the stakeholders are, what they asked for, and how each analytical decision maps back to a real business objective. It is a living document, updated as the project progresses through cleaning, analysis, and delivery.

1. Project Overview

A simulated data warehouse environment for a food delivery platform, built on real order data (Delhi NCR, Sep 2024–Jan 2025, 21,321 orders across 6 restaurant brands / 19 outlets) with a synthetic delivery-operations layer layered on top to enable partner-level SLA analysis, which the source data does not natively support. Two departmental stakeholders have each brought a distinct business objective; both require the same foundational data cleaning before any analysis can be trusted.

2. Stakeholder — Priya Nair

Title	Head of Customer Retention & Growth
Department	Growth / Retention Marketing
Reports to	VP of Marketing
Primary tension	Heavy reliance on discounts to drive orders, at the cost of unclear long-term loyalty

Business Objective (OKR)

> **Objective:** Build a customer base that keeps ordering because they love the service — not because of the next discount code.

- **KR1:** Establish baseline — what % of order volume and revenue comes from customers who only order during active promo periods (no existing baseline; first pass required).
- **KR2:** Lift 90-day repeat order rate among non-promo-dependent customers (target set only after KR1 baseline is known).
- **KR3:** Reduce discount-dependent order volume as a share of total, without reducing total order volume (guards against a naive “just cut discounts” fix).

Why this matters / industry grounding

Food delivery platforms commonly lean on discounting to acquire and retain customers, which risks training customers to expect discounts and eroding full-price retention over time — a documented industry pattern this project treats as the working hypothesis to test against real order data.

Initial analysis approach

Cohort retention first (are non-promo customers actually stickier over time, or does everyone eventually churn?), **then RFM** to segment today's customer base by promo-dependency. Cohort answers the trend question; RFM answers who matters right now.

Findings & Recommendation

Finding 1 — Discount status at first order does not predict retention. Comparing customers whose first order used a discount vs. full price, across four monthly cohorts (Sep–Dec 2024), retention gaps were only 1–5pp and inconsistent in direction (flipped in the December cohort), indicating no meaningful causal relationship. An earlier version using lifetime discount % showed a large, consistent gap, but was discarded after being traced to a look-ahead bias (using future orders to explain past retention) — this corrected version replaces it.

Finding 2 — Promo-dependent revenue share is real but threshold-sensitive. Using a >50% lifetime-discount-usage definition, 45.27% of total revenue comes from promo-dependent customers, but re-testing at 40/60/70% thresholds moved that figure between 37.72% and 52.64%, a 15pp swing. The defensible finding is therefore a range — roughly 38–53% of revenue sits with heavy discount users — not a single precise percentage.

Finding 3 — That revenue is concentrated in a small high-spend group, not evenly spread. Within the promo-dependent segment, the top 10% of customers by spend generate 30.78% of the segment's total revenue, and the top 20% generate 46.6%, while the bottom 10% contributes only 2.34%. The "45% of revenue is promo-dependent" headline is really driven by a relatively small number of high-value customers, not a broad habit across the customer base — a targeted, not blanket, response is warranted.

Recommendation: do not pursue blanket discount reduction. Investigate top-decile promo-dependent customers directly; treat one-time customers (37% of revenue) as a distinct, higher-priority problem; reframe KR1's target-setting around the 38–53% range rather than a single point estimate.

Finding — Cohort retention drops off steeply and immediately

Across all five monthly cohorts (Sep 2024–Jan 2025), retention falls from 100% at month 0 to roughly 20–26% by month 1 alone — the September cohort shows 100% → 25.59% → 21.15% → 19.22% → 15.91% over four months. The steepest drop happens in the very first month after a customer's initial order, with the curve flattening afterward rather than declining steadily — consistent across every cohort observed, meaning this is a structural feature of the business, not a one-off blip.

Recommendation: concentrate retention efforts on the immediate post-first-order window (e.g. a follow-up nudge within days of order 1), since that's where the overwhelming majority of customer loss happens — later-month interventions have a much smaller pool of at-risk customers left to act on.

Finding — RFM segmentation identifies four actionable customer groups

Segmenting all 11,607 customers on Recency, Frequency (fixed bands after quintile scoring broke down — 66% of customers tied at frequency=1), and Monetary (quintile-scored) produces: Champions ~394 (3.4%, recent + frequent + high-spend), Loyal ~1,340 (11.5%), At-risk 247 (2.1%, previously valuable but gone cold), Churned ~714 (6.1%), and Other ~8,910 (76.7%). Segments were spot-checked against real customer records and confirmed to match their labels.

Recommendation: hand the Champions list to the loyalty team, the At-risk list for targeted win-back (justified by proven prior value), and treat Churned as lower-priority/automated outreach only.

Addendum — RFM segment comparability across cohorts

Cross-tabulating RFM segment against cohort_month showed “Other” rising steadily from 56.88% (Sep 2024 cohort) to 89.64% (Jan 2025 cohort), while Churned/At-risk shares fell in the same direction (Sep: 17.94% Churned → Jan: ~0%). Checking average R/F/M scores by cohort confirmed the mechanical cause: avg_r_score climbs from 2.09 (Sep) to 4.75 (Jan) purely because newer customers cannot have ordered long ago, while avg_f_score and avg_m_score fall in parallel because newer cohorts haven't had time to accumulate orders or spend — a combination that doesn't match any named segment rule by construction.

Implication: the Churned/At-risk-concentrate-in-older-cohorts finding is confirmed as an expected, stable pattern, not an accelerating problem — but RFM segment shares should never be compared directly across cohorts of different ages without this context. A January customer labeled “Other” is not less valuable than a September customer labeled “Loyal” — they simply haven't had the calendar time to be classified as anything else yet.

3. Ongoing KPIs — Priya (Retention & Growth)

Standing caveat — read this before any KPI below: any metric in this dashboard that depends on elapsed time since a customer's first order (repeat rate, one-time-customer share, RFM-derived segments, At-risk/Churned counts) will look artificially worse for younger cohorts — not because those customers behave differently, but because they haven't had the calendar time to be fairly measured against a fixed window yet. This is right-censoring, the same failure mode identified across three separate KPIs below independently. Never compare a young cohort's numbers directly against an older cohort's without first checking whether both have had equal time to mature.

#	KPI	Cadence	Red flag direction
1	Repeat-order rate within 30 days of first order	Monthly, per cohort	Falling (adjust for censoring)
2	Month-1 retention rate	Monthly, per cohort	Below ~20%
3	% of revenue from one-time customers	Monthly, per cohort	Rising (adjust for censoring)
4	Discount-dependent revenue share (fixed 50% threshold)	Monthly	Rising trend
5	New Champions added per month	Monthly (flow)	Falling
6	At-risk count, cohorts ≥ 2 months old only	Monthly	Rising
7	Complaint rate per 100 orders	Monthly	Rising

KPI 1 result — Repeat-order rate within 30 days (by cohort)

Declines steadily from 30.42% (Sep 2024) to 14.96% (Jan 2025). Very likely a right-censoring artifact — January customers had as little as a few days, not a full 30, to register a repeat order before the dataset's max date. Should not be presented as a real behavioral decline without this caveat; recalculate excluding cohort-months without a full 30 days of observable history.

KPI 3 result — One-time-customer revenue share (by cohort)

Climbs steeply from 17.93% (Sep 2024) to 72.02% (Jan 2025) — the same right-censoring mechanism as KPI 1. The original static 37% one-time-revenue baseline (Priya KR1) was a blend of mature, mostly-converted cohorts and still-censored young ones, masking real variation. Only report/act on cohort-months with 2+ months of history.

KPI 2 result — Month-1 retention rate (by cohort)

Declines from 25.59% (Sep 2024) to 15.36% (Dec 2024). Unlike KPIs 1, 3, and 5, this metric uses calendar-month buckets where every cohort shown has a fully-observed month-1 window within the data range (Jan cohort correctly omitted rather than shown at 0%, since its month-1 would need unavailable February data). This decline is NOT a censoring artifact — it is the most trustworthy trend signal in the KPI set, precisely because it is structurally immune to the issue affecting the others.

Recommendation: prioritize root-cause investigation into what changed operationally between September and December that could explain a genuine ~40% relative decline in month-1 retention over three months — new customer mix, onboarding experience, or pricing/promo changes are candidates worth checking first.

KPI 4 result — Discount-dependent revenue share (monthly, order-level)

Holds in a stable 65-80% band across all five months (Sep 79.96%, Oct 75.42%, Nov 77.09%, Dec 65.01%, Jan 71.69%) with no clear directional trend. This KPI is order-level and month-scoped rather than customer-lifetime-scoped, so it requires no maturity window and is safe to compare directly across months — the one KPI in this set structurally immune to right-censoring by design, not just by coincidence. December's dip to 65.01% (the low point in an otherwise consistent band) is worth a targeted follow-up by subzone/restaurant.

Recommendation: track as the primary trustworthy month-over-month promo-dependency signal; flag any month falling outside the observed 65-80% band for investigation.

KPI 5 result — New Champions conversion rate (by cohort)

Declines steeply even after controlling for shrinking cohort size (7.33% Sep → 2.82% Oct → 2.15% Nov → 1.64% Dec), ruling out “smaller cohorts” as the primary explanation. However, Champions status requires 5+ orders AND top-quintile spend AND high recency — criteria that take real elapsed time to accumulate. Only September has had the full 5-month window; Oct/Nov/Dec are all still time-constrained to varying degrees, so their lower rates are still substantially explained by insufficient time to mature, not necessarily lower underlying customer quality.

Recommendation: do not compare monthly Champion-conversion rates directly until each cohort has had a comparable maturity window (likely 4-5+ months). Treat current Oct-Dec figures as provisional floors, not final quality readings.

KPI 6 result — At-risk count, mature cohorts only (2+ months old)

Restricting to cohorts with ≥ 2 months of history (Sep/Oct/Nov) produces 250 customers, closely matching the unrestricted total of 247 — confirming At-risk was already naturally concentrated in older cohorts by definition (it requires low recency), so the maturity restriction has minimal practical effect. Breakdown: September 190

(76%), October 58, November 2, December/January 0 (excluded as immature).

Recommendation: track by cohort_month, not as a single total — the concentration itself (which cohort is driving At-risk) is the actionable insight for targeting win-back campaigns.

Dataset Extension — 12-month data (client shared 7 additional months)

The original 5-month dataset (Sep 2024–Jan 2025) was extended with 7 simulated additional months (Feb–Aug 2025) to reach 12 months total, mirroring a client sharing more data on request. The extension used bootstrap-sampled rows (preserving real joint attribute relationships) and, critically, repeat-customer behavior driven by the actual measured retention curve (~19% month-1 reorder probability, ~82% month-over-month persistence thereafter) rather than random generation — documented in build_data_extension.py / build_ops_extension.py.

Re-testing KPI 1 (30-day repeat rate) on the extended data corrected the original right-censoring diagnosis: Sep–Dec 2024 are exactly unchanged from the original figures, since none of those cohorts' 30-day windows actually extended past the original data's edge — meaning the Sep→Dec decline was real behavior all along, not a censoring artifact. Only January was genuinely censored before (14.96% → 22.69% once resolved).

The extension also revealed its own limitation: new Feb–Jul cohorts show an artificially uniform ~9-10% band on KPI 1 specifically, because the simulation decided repeat behavior at calendar-month granularity while placing the simulated order on a random day within that month — frequently missing a strict 30-day rolling window. KPI 2 (month-1 retention, calendar-month granularity) was re-tested and confirmed unaffected — Sep–Dec unchanged, January resolved (17.60%), Feb–Jul show a smooth, plausible 18-20% band.

KPI	Granularity	Extended-data status
1. 30-day repeat rate	Day-level rolling	NOT reliable past Jan 2025 — use original 5-month window only
2. Month-1 retention	Calendar-month	Confirmed safe
3. % revenue, one-time customers	Lifetime count	Safe
4. Discount-dependent revenue share	Monthly aggregate	Safe
5. New Champions rate	RFM (R/F/M)	Safe from this artifact; minor spend-variation caveat
6. At-risk, mature cohorts	RFM + cohort age	Safe from this artifact; same caveat as #5
7. Complaint rate	Monthly aggregate	Safe

General principle: any KPI whose granularity matches the simulation's generative logic (calendar-month decisions) is trustworthy on the extended data; any KPI requiring finer-than-month timing precision is not, unless the simulation is rebuilt with day-level repeat-timing logic.

4. Stakeholder — Arjun Mehta

Title	Head of Delivery Operations
Department	Operations / Logistics
Reports to	VP of Operations

Primary tension	Inconsistent on-time delivery performance; unclear whether root cause is partners, restaurants, or sp

Business Objective (OKR) — framed as improvement projects

Objective: Make on-time delivery a promise customers can actually rely on — not a coin flip.

- **Project 1 (KR1):** Root-cause SLA misses — identify whether they concentrate in specific zones, delivery partners, or restaurant prep time. Problem definition first, no assumed cause.
- **Project 2 (KR2):** Reduce late-delivery rate by a stated percentage-point threshold in the worst-performing segment identified in Project 1 — baseline measured first, impact tracked before vs. after.
- **Project 3 (KR3):** Detect and flag delivery partners whose performance metrics are statistically inconsistent with platform norms — an escalation/vendor-performance problem, not a soft target.

Interview framing note — Workwize Operations Project Manager role

This stakeholder's workstream is deliberately structured to mirror an Ops PM / Service Operations role (e.g. Workwize's Retrievals/Delivery/Warehousing service lines): root cause → solution → measured outcome, vendor (delivery partner) performance management, and before/after impact validation. Each project below will be written up with a stated baseline, the change tested, and the measured after-state — directly answering “track performance before and after changes to validate impact.” Ambiguous judgment calls made along the way (e.g. establishing table grain, resolving a non-deterministic dedup tie-break) are logged as evidence of comfort operating in ambiguity, not just clean final answers.

Why this matters / industry grounding

Delivery time accuracy and order completeness are commonly cited as key drivers of customer loyalty on food delivery platforms — operational reliability is treated here as directly tied to retention, not a purely internal cost metric.

Initial analysis approach

Funnel / stage breakdown first (prep time vs. pickup wait vs. transit — three different owners, three different fixes), **then outlier/threshold flagging** for the partner-integrity question in KR3.

Data note: the source export has no delivery-partner identity and no actual delivery timestamp. The **delivery_partners_dim** and **delivery_ops_fact** tables are fully synthetic constructions, documented in the project's Data Dictionary, built specifically to make this stakeholder's questions answerable in a way the raw data alone does not support.

Finding — Project 1: Root-cause breakdown by restaurant, zone, and partner

Two problem restaurant brands (tandoori junction, dilli burger adda) show a structurally elevated late rate (14-25%) across every outlet, vs. 5-11% for other brands — a brand-wide, not single-outlet, issue. Four of 50 delivery partners run late rates of 39.55-50.59%, roughly 4-5x the clean platform baseline. Several zone-level rates that looked extreme in raw form were volume artifacts (e.g. a single delivery showing 100% late) and were flagged rather than reported at face value.

Finding — Project 2: Counterfactual impact of fixing the two root causes

Simulating a fix for the 4 flagged partners (bringing their late rate to the clean 8.21% baseline) projects the platform-wide late rate falling from 9.49% to 8.31%, a 1.18pp improvement. The same exercise for the two problem restaurant brands shows their clean-baseline rate (9.36%) barely differs from the actual platform rate (9.49%), because those brands represent under 2% of total order volume — fixing them would not move the platform number meaningfully despite their individually bad rate.

Recommendation: prioritize the partner-level fix as the primary KR2 lever; track the restaurant-brand issue as a separate, lower-priority service-quality initiative rather than folding it into the same target.

Finding — Project 3: Partner outlier detection, verified robust

Using a 2-standard-deviation threshold (mean 11.11%, stddev 10.24%), 4 partners were flagged with late rates 4-5x typical. Since extreme outliers can inflate their own detection threshold, the baseline was recomputed excluding these four (clean mean 8.21%, stddev 2.21%) — all four still exceeded the stricter threshold (12.63%) by a wide margin, confirming the finding is not a detection-method artifact.

Recommendation: escalate the 4 partners for individual performance review; re-run this detection quarterly as new data accumulates.

Finding — Zone/distance confound: DLF Phase 1 has a genuine, unexplained short-distance issue

Comparing late rate within the same distance band (controlling for distance) shows DLF Phase 1 at 26.07% late for deliveries ≤1km, vs. 8-15% for every other zone at the same distance — a real zone effect, not a distance artifact, and one that shrinks sharply at longer distances. Sector 4 — the other zone named in the original escalation — shows no such pattern at any distance, meaning its escalations are explained by distance/volume, not a genuine zone issue.

Three candidate explanations were tested: restaurant prep time (KPT) is elevated only ~1-1.5 minutes in DLF Phase 1's short band vs. its own other bands; rider pickup wait is elevated only ~0.7-0.8 minutes; partner assignment mix was ruled out entirely (the 4 known bad-apple partners are not overrepresented in this zone/band — two have zero deliveries there). Combined, KPT and rider wait account for roughly 2-2.5 minutes of delay against a 15-20pp late-rate gap — a small fraction of the effect. Traffic/road conditions remain plausible but are structurally untestable, since the synthetic transit-time model uses a distance-only formula with no zone-specific term.

Recommendation: escalate DLF Phase 1's short-distance performance for on-the-ground investigation (physical route/traffic audit) — the correct, honest limit of what this data can resolve. Separately, one partner (DP0041) was found to handle 39.69% of this specific zone/band's deliveries — a concentration risk worth flagging on its own, unrelated to the lateness question.

5. Ongoing KPIs — Arjun (Delivery Operations)

#	KPI	Cadence	Threshold / red flag
1	Platform late rate, 7-day rolling avg	Daily	Sharp spike — catch problems same-day
2	Platform late rate, 3-month rolling avg	Monthly	Sustained upward drift
3	DLF Phase 1, ≤1km late rate	Monthly	Not returning to 8-15% peer band

4	Partner outlier re-detection (2-SD, per-month baseline)	Monthly	Any partner newly crossing threshold
5	Restaurant brand late rate (relative volume flag, 30% of own avg)	Monthly	Outside brand's established band
6	Partner concentration risk (top partner's % per subzone)	Monthly	>30% — build a second partner in that zone

KPI 1 & 2 results — Rolling late rate (7-day and 3-month)

Both built on deduped delivery data (order_id, partner_id grain correctly collapsed) with raw counts rolled up before computing a single rate, not an average of pre-computed daily/monthly rates. The 3-month view shows a stable band (~8.9-10.6%) across all 12 months, dipping slightly in winter/spring and rising slightly through summer — no runaway trend. Both KPIs are live, daily/monthly-refreshable signals for "is something wrong right now" vs. "is this a sustained drift."

KPI 3 result — DLF Phase 1 short-distance tracker

Tracked monthly across the full 12 months, late rate improved substantially from its worst point (34.55% Sep 2024) to its best (12.24% Jul 2025), but relapsed to 25.58% as recently as February 2025 and has never sustainably dropped into the 8-15% band every other zone shows at this distance. Read as “meaningfully better, not resolved” — continue tracking rather than treating as closed.

KPI 4 result — Monthly partner outlier re-detection

Re-running 2-SD outlier detection independently each month (fresh mean/stddev per month, not the original baseline) flagged 45 partner-months total across 12 months — every single one belonging to the original 4 bad-apple partners (DP0004/46/48/50). Zero new partners emerged, zero false positives. This is a stronger validation than the original one-time finding: anomalous independently in 12 separate monthly tests, not just once.

Recommendation: 12 consecutive months of consistent failure rules out a temporary bad patch — these 4 partners need direct intervention, not continued monitoring.

KPI 5 result — Restaurant brand late rate (relative volume threshold)

A fixed volume threshold was rejected after checking the distribution (brand monthly volumes ranged 1-3,016, a 17x mean/median gap) — instead, each brand is flagged low-volume only below 30% of its own 12-month average. Tandoori junction and dilli burger adda both show late rates persistently in the mid-teens to high-20s across nearly every month, confirming the brand-wide issue has not resolved over the full 12 months, unlike DLF Phase 1's partial recovery.

KPI 6 result — Partner concentration risk

DP0041 is the single largest partner in 70+ of 76 subzone-month combinations across 12 months, consistently at 30-45% of orders in every substantive zone — exceeding Arjun's own 30% risk threshold almost every month, almost everywhere. This is a materially larger finding than the original single-slice discovery suggested: not an isolated pattern, but a structural, platform-wide resilience risk sustained for a full year.

Recommendation: the single highest-urgency finding across all of Arjun's KPIs. Warrants an active second-partner build-out program per zone with a stated remediation target (e.g. “no partner above 30% in any zone within 6 months”), not passive monitoring.

6. Amazon Leadership Principles — Applied in This Project

This project is deliberately run the way an analyst would operate inside an organization that holds itself to these principles — not as a slogan, but as a working discipline reflected in specific decisions made during the project. Each principle below is paired with a concrete example already produced in this project, so it can be cited directly in an interview.

Principle	How it showed up in this project
Customer Obsession	Arjun's SLA definition was built distance-based rather than a flat number, because a one-size promise c
Ownership	When a data quality bug was found (mismatched duplicate KPT values, a leaked answer-key column), i
Insist on the Highest Standards	Every cleaning step was verified against an actual row count or distinct-value count before being accep
Dive Deep	The KPT null count (711 vs. 709) discrepancy was traced down to a single root cause — non-determinis
Are Right, A Lot	The delivery_ops_fact dedup decision was revisited after the first assumption (partition by order_id alon
Bias for Action	Diagnostics were run issue-by-issue in a live worksheet rather than over-planning a perfect query up fro
Frugality	Snowflake trial credits and a single XSMALL warehouse were used deliberately, with BigQuery's free Sa
Invent and Simplify	A single reproducible Python script generates all synthetic tables and injected data-quality issues, rathe
Deliver Results	Every analytical step is tied back to a stated Key Result for a named stakeholder, not run as an unattac

This section will be updated as the project reaches the analysis and dashboard phases — additional rows will be added as more principles surface naturally in later decisions (e.g. Earn Trust, Think Big) rather than being forced in retroactively.

7. Current Project Status

Stage	Status
Dataset extension (12 months, Sep 2024-Aug 2025)	Complete — 7 simulated months added via retention-curve-driven
Raw data loaded to Snowflake (4 tables)	Complete
stg_orders (staging layer)	Logic complete and verified (21,321 rows, 709 KPT-missing flags)
stg_delivery_ops (staging layer)	Logic complete and verified (21,350 rows). Rebuilding on new Sno
dbt project	Paused — Snowflake trial access is the current constraint, not dbt
Cohort / RFM analysis (Priya)	Complete — diagnostics, RFM segmentation, cross-cohort addend
Root-cause + partner outlier + impact projection (Arjun)	Complete — Projects 1-3 done (restaurant/zone/partner root caus
Zone/distance confound analysis (Arjun)	Complete — DLF Phase 1 confirmed as a genuine short-distance
Ongoing KPIs (Arjun)	Complete — 6 KPIs built: rolling late rate (7-day/3-month), DLF Ph

GitHub repository with KPI preview visuals	Planned
Dashboard tool priority	1) Power BI (live-connected, Windows) → 2) Tableau paid trial →

Document version 1.0 — generated at project setup. Will be revised as stages complete.

8. Interview Prep — Operational Analytics Questions

Logged separately from the hands-on project so it doesn't mix with the Priya/Arjun analysis. To be worked through as its own exercise, same hint-first coaching style. Source: practice question set styled after Swiggy-type operational analytics interviews.

- How would you measure the health of Swiggy's delivery experience using data?
- Write a SQL query to find the top 5 restaurants by repeat order rate in each city.
- If average delivery time increased by 8 minutes last week, how would you investigate why?
- Design a dashboard for a city operations manager. What metrics would you include?
- How would you calculate customer lifetime value for a food delivery app?
- Write a query to find customers who ordered 3+ times in their first month but never again.
- How would you A/B test a new app feature to measure impact on order frequency?
- If a promo campaign drove 40% more orders but profit dropped, what's your analysis?
- Write a query to rank delivery partners by on-time delivery rate per zone.

Status: not yet attempted. Several overlap directly with work already done in this project (repeat-order-rate → RFM/frequency work; partner ranking per zone → Arjun's Project 1/3; promo-margin tradeoff → Priya's promo-dependency findings) — worth explicitly drawing that connection when answering, rather than starting from scratch.

9. Analysis Steps — General Framework

The reusable sequence followed for any stakeholder engagement, from a raw ask to a delivered recommendation. Applies regardless of which stakeholder or dataset is involved.

1. Clarify the objective (OKR) — Establish what decision this analysis will drive and what north-star metric matters, before touching any data.

2. Data audit and cleaning — Row counts, nulls, duplicates, orphaned keys, impossible values. Flag data quality issues rather than silently deleting or fixing them.

3. Initial diagnostic, matched to objective type — Pick the technique the question actually calls for — cohort for a trend question, RFM for a current-value question, funnel/dimension breakdown for “where does it break,” outlier detection for “is someone gaming it.” Don't default to a favorite technique.

4. Cross-reference diagnostics against each other — Two diagnostics run separately can still hide a relationship between them — e.g. does a trend-level pattern concentrate inside one segment identified separately?

5. Root-cause and deep-dive — Let the initial finding generate the next hypothesis; test it with joins, window functions, or before/after comparisons.

6. Quantify and stress-test — Turn a hunch into a number with a stated decision threshold, then check whether that number survives scrutiny — threshold sensitivity, concentration/outlier checks, look-ahead bias, robustness to the outliers used to detect it.

7. Recommendation tied back to the objective — Every finding connects to a stated lever and its cost/effort tradeoff — not a number floating without an action attached.

8. Define ongoing KPIs — A small set (5-8) of metrics the stakeholder's team tracks continuously after the one-time analysis ends — the dashboard's north-star layer.

9. Build the dashboard — Self-serve access to the numbers, so the stakeholder isn't dependent on a fresh query every time a question comes up.

10. This Project — Steps Taken, In Order

A concrete log of what's been executed on this specific project, in sequence, mapped against the general framework above.

- Sourced and profiled real order data (21,321 orders, Delhi NCR, Sep 2024-Jan 2025); documented what was real vs. synthetic
- Injected realistic warehouse mess (dedup, casing/whitespace, mixed date formats, orphaned FKs, impossible values, nulls) via a reproducible Python script
- Set up Snowflake (database, schemas, warehouse) and loaded 4 raw tables
- Built and verified stg_orders: dedup with deterministic tiebreak, casing/whitespace fix, date parsing, orphan flag, impossible-value flags, distance_km conversion
- Built and verified stg_delivery_ops: established (order_id, partner_id) grain to correctly preserve legitimate reassignment rows while removing the one true duplicate

- Priya — Step 3: Cohort retention analysis (trend question) — found steep, immediate month-1 drop-off
- Priya — Step 3: RFM segmentation (current-value question) — Champions/Loyal/At-risk/Churned, fixed frequency bands after quintile scoring broke down
- Priya — Step 5/6: Promo-dependency deep dive — caught and corrected a look-ahead bias, ran threshold-sensitivity and revenue-concentration checks
- Arjun — Step 3: Root-cause breakdown by restaurant, zone, and partner (Project 1)
- Arjun — Step 6: Counterfactual impact projection for fixing bad-apple partners vs. problem restaurants (Project 2) — found partner fix has real leverage, restaurant fix does not (low volume)
- Arjun — Step 5/6: Partner outlier detection via 2-SD threshold, re-verified against an outlier-free baseline (Project 3)
- Priya — Step 4: Cross-referenced RFM segment against cohort_month; found “Other” climb (56.88%→89.64%) is an RFM-mechanics artifact, not a real trend
- Priya — Step 8: Built and documented all 7 ongoing KPIs, distinguishing which are right-censoring-vulnerable (1, 3, 5, 6) vs. structurally immune (2, 4) — month-1 retention decline confirmed as a genuine trend, not an artifact
- Arjun — Zone vs. distance confound check: DLF Phase 1 confirmed as genuine short-distance issue; KPT, rider wait, partner-mix tested; escalated as physical/logistical investigation
- Arjun — Step 8: Built and documented 6 ongoing KPIs on the full 12-month dataset, including monthly partner re-detection (validated the 4 bad-apple partners across 12 independent monthly tests) and partner concentration risk (surfaced DP0041 as a platform-wide, sustained resilience risk — the single highest-urgency finding in the project) — BOTH STAKEHOLDERS' CORE ANALYSIS AND KPI WORK NOW COMPLETE
- [PLANNED] Step 7/8 for both stakeholders: final recommendations and ongoing KPI definitions
- [PLANNED] GitHub repo with preview visuals; Power BI live-connected dashboard (Tableau as fallback)